SEVI: Safe Evidential Real-World Reinforcement Learning

***Abstract*—Real-world reinforcement learning has enabled contact-rich manipulation tasks to be trained directly on physical robots, but unsafe interactions during online learning remain a key barrier to practical deployment. We propose Safe Evidential Real-World Reinforcement Learning (SEVI), a framework for reliability-aware safety intervention during real-world policy learning. SEVI introduces an Evidential Safety Critic that estimates both future constraint-violation risk and epistemic uncertainty in a single forward pass. During online learning, SEVI evaluates each task-policy action before execution; if the action is predicted to be unsafe or the prediction is insufficiently supported by evidence, SEVI overrides it with an offline-trained recovery policy that guides the robot toward a lower-risk region. We evaluate SEVI on two simulated manipulation tasks and a real-world USB insertion task under in-distribution and distribution-shifted conditions. Across task-learning and safety-critic baselines, SEVI achieves the strongest trajectory-level safety prediction, the highest final task success, and the lowest unsafe-event counts. SEVI also reduces unsafe events by 13.4–57.1% relative to recovery-based safe RL while preserving low-latency execution-time filtering. The supplementary video is available at this double-anonymous link: `Supplementary Video`.**

***Index Terms*—Safe RL, Manipulation, Safety, Uncertainty.**

I. Introduction

Reinforcement learning (RL) holds the promise of enabling robots to acquire complex manipulation skills through trial and error. Recent real-world RL systems have shown that contact-rich manipulation tasks can be trained directly on hardware within a practical time budget. These systems typically rely on demonstrations, sample-efficient off-policy learning, learned rewards, and carefully designed controllers [1], [2]. However, training RL on physical robots remains challenging because exploration can lead to unsafe interactions, such as excessive contact forces, object damage, workspace violations, or failures requiring human intervention. Therefore, as real-world RL becomes increasingly sample-efficient, a central remaining challenge is to reduce unsafe interactions during online training.

This challenge is particularly pronounced in complex tasks, where safety depends on perception, object geometry, contact dynamics, and the evolving task policy [3]. Existing real-world RL pipelines provide an effective foundation for online learning with prior data and off-policy updates. However, they do not explicitly address the autonomous intervention problem: *determining when a proposed task action should be rejected and replaced by safe recovery behavior*. One way to address this intervention problem is to augment the task policy with a recovery mechanism. Recovery-based safe RL methods instantiate this idea by switching to a recovery policy when a learned safety critic predicts future constraint violations [4]. A key limitation is that these intervention decisions rely on the reliability of the learned safety estimate, which may degrade under partial observations or in state-action regions poorly represented in prior data.

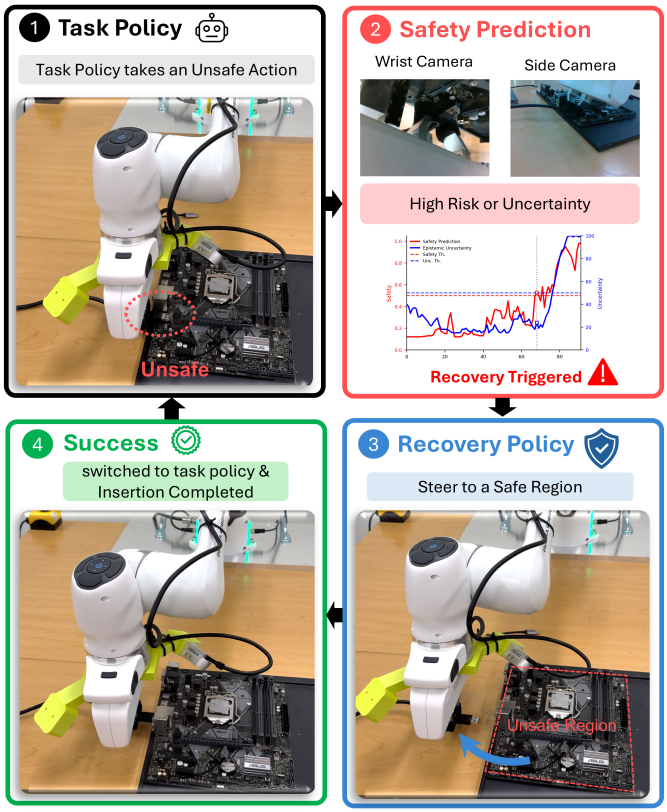


Fig. 1: Illustration of SEVI on the real-world USB insertion task. When the task policy attempts an unsafe insertion, SEVI detects elevated safety risk or epistemic uncertainty and triggers the recovery policy. The recovery policy steers the robot toward a safer region before returning control to the task policy for continued online RL training.

Existing approaches to uncertainty-aware safety attempt to detect unreliable predictions under distribution shift [5]. World-model-based safety filters use learned latent dynamics to imagine future outcomes and identify potential out-of-distribution (OOD) failures [6]. Although this is a promising direction, it relies on accurate long-horizon prediction and sufficient coverage of safety-critical interactions. In complex manipulation, limited unsafe data and partial observability can cause learned dynamics to produce optimistic predictions in unfamiliar regions, leading unsafe actions to be classified as safe. Although sampling-based uncertainty methods can improve reliability estimation, ensembles [7] and Monte Carlo dropout [8] require multiple models or repeated forward passes, which increases latency for real-time robot control [9].

To address this problem, we propose Safe Evidential Real-World Reinforcement Learning (SEVI), a framework for uncertainty-aware autonomous intervention during online real-world RL. SEVI introduces an Evidential Safety Critic (ESC), which models safety prediction as evidence accumulation and estimates both future constraint-violation risk and epistemic uncertainty in a single forward pass. Unlike sampling-based

uncertainty methods, the ESC does not require ensembles or repeated stochastic inference, making it suitable for low-latency safety decisions. During online training, SEVI evaluates each action proposed by the task policy using both predicted risk and prediction reliability. If the action is predicted to be unsafe or the prediction is insufficiently supported by evidence, SEVI overrides the task action with an offline-trained recovery policy that guides the robot toward a lower-risk region as illustrated in Fig. 1. This intervention mechanism reduces unsafe interactions without modifying the underlying off-policy RL objective.

We evaluate SEVI on two simulated manipulation tasks and a real-world USB insertion task with a Franka robot. The experiments show that SEVI improves safety prediction under distribution shift, provides better-calibrated uncertainty than dropout and ensemble baselines, and reduces unsafe events during online real-world RL while maintaining strong task performance. In summary, this paper makes the following contributions:

- We introduce a modular real-world RL safety architecture that decouples task-policy learning from execution-time safety intervention while preserving the underlying task-learning objective.
- We propose an Evidential Safety Critic that estimates both future constraint-violation risk and epistemic uncertainty in a single forward pass, enabling low-latency reliability-aware safety assessment.
- We integrate evidential safety prediction with an offline-trained recovery policy to enable autonomous intervention during online real-world RL with prior data, while remaining compatible with standard off-policy RL pipelines.

II. Related Work

Real-world RL. Efficient learning from limited physical interaction remains challenging in the presence of high-dimensional observations, sparse rewards, and practical reset constraints. Recent progress has combined demonstrations or prior data with sample-efficient off-policy RL, enabling policies to be trained directly on physical robots within practical time budgets [1]. Sample Efficient Robotic Reinforcement Learning (SERL) provides a full-stack real-world RL pipeline that integrates robot control, reward inference, asynchronous training, and Reinforcement Learning with Prior Data (RLPD) off-policy updates for manipulation tasks [2]. These systems demonstrate that real-world RL can be practical when algorithmic and system-level components are carefully integrated. For complex manipulation tasks, online training can still require long hardware sessions with continuous human monitoring, resets, or corrective intervention. This supervision burden limits scalability, especially when unsafe behavior can occur during exploration. SEVI targets this bottleneck by adding an autonomous safety layer that rejects risky or unreliable task actions and invokes recovery during online training.

Safe RL. Classical safe RL commonly formulates safety through constrained policy optimization [10], while control-theoretic approaches use control barrier functions or Hamilton–Jacobi reachability to enforce safety [11], [12]. Predictive and latent safety filters extend safety reasoning to learned models and high-dimensional observations [6]. Such approaches often rely on predefined constraints, dynamics models, or explicit safety specifications. Recovery-based methods are more closely related to SEVI, as they override risky task actions before unsafe states are reached [4]. However, these methods typically trigger intervention from predicted risk alone, which may become unreliable under partial observations, policy-induced distribution shift, or limited data coverage. SEVI additionally considers the reliability of the safety estimate when determining whether recovery is required.

Uncertainty and Failure Prediction. Uncertainty estimation is widely used to identify unreliable predictions in learned robotic systems. Monte Carlo dropout and deep ensembles estimate epistemic uncertainty through stochastic inference or model disagreement [7], [8]. These methods are effective, but they require repeated forward passes or multiple models, which increases latency for real-time robot control. Recent failure-detection methods monitor deviations from successful behavior. For example, FAIL-Detect formulates runtime failure detection as sequential OOD detection and uses conformal prediction to detect failures from successful rollouts alone [13]. Such methods improve runtime awareness, but they mainly serve as monitors and do not directly provide autonomous intervention for online RL. Evidential learning provides a sampling-free alternative for uncertainty estimation. Instead of sampling over network weights or model instances, it models prediction as evidence accumulation and estimates uncertainty from the parameters of an evidential distribution [9]. Building on this formulation, our Evidential Safety Critic predicts both future violation risk and epistemic uncertainty in a single forward pass. This design connects uncertainty estimation to action selection: recovery is triggered when an action is either high-risk or insufficiently supported by evidence.

III. Problem formulation

We consider safe real-world RL for safety-critical robotic manipulation, where a robot learns a policy through physical interaction while minimizing unsafe events during training. We model the task-learning problem as a Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \rho, \mathcal{P}, r, \gamma)$,where $s \in \mathcal{S}$ is the state observation (e.g. image observation with the robot's proprioceptive state information), $a \in \mathcal{A}$ denotes the control action (e.g. the desired end effector pose), $\rho(s_o)$ is the distribution over initial states, $\mathcal{P}$ is the unknown and potentially stochastic transition probabilities that depend on the system dynamics, $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function which encodes the task and $\gamma \in [0, 1)$ is the discounted factor.

$$J_{\text{task}}(\pi_{\text{task}}) = \mathbb{E}_{\pi_{\text{task}}, P}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right]. \tag{1}$$

Directly optimizing this objective on hardware can lead to unsafe robot-object interactions that require intervention. To model such events, we introduce a binary constraint signal $c_t \in \{0, 1\}$, where $c_t = 1$ indicates a safety violation and

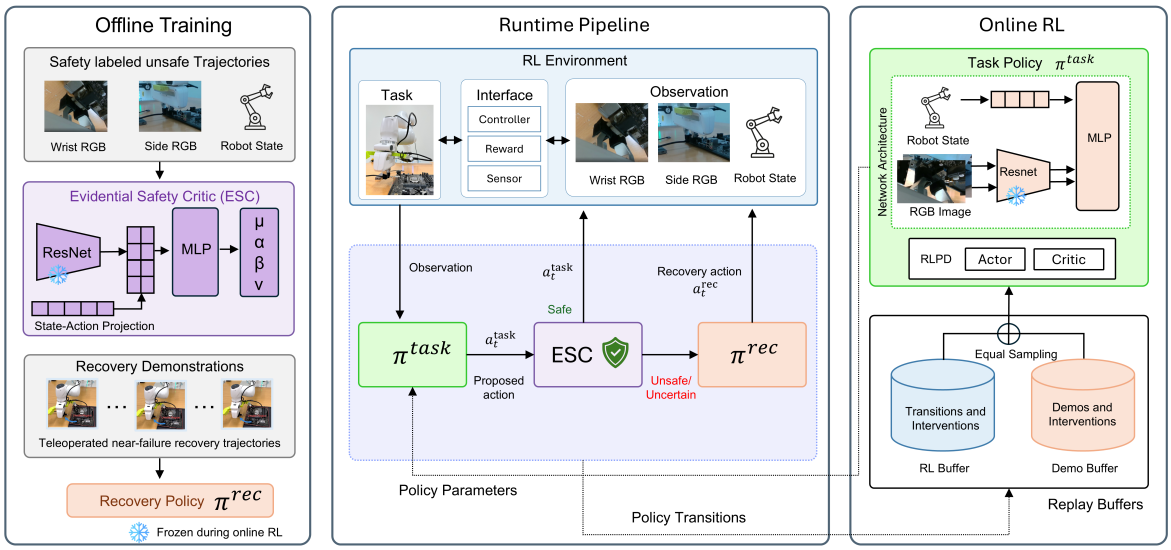


Fig. 2: **Overview of the SEVI framework**. In offline training, failure and recovery demonstrations are used to train the Evidential Safety Critic and recovery policy. During the runtime pipeline, the ESC evaluates task-policy actions using predicted risk and epistemic uncertainty and switches to the recovery policy when needed. In online RL, RLPD updates the task policy using executed transitions from the RL buffer together with prior demonstrations from the demo buffer.

$c_t = 0$ otherwise. In simulation, this signal can be obtained from privileged task information, such as object toppling, collision, or workspace violation. On hardware, it is obtained from task-specific safety observations, such as force thresholds, keep-out regions, or failure annotations. Following prior safety-critic formulations [4], we define the risk value function as the discounted probability of future constraint violation after executing action a_t in state s_t and then following policy π:

$$Q^{\pi}_{\text{risk}}(s_t, a_t) = \mathbb{E}_{\pi,P}\left[\sum_{k=0}^{\infty} \gamma^k_{\text{risk}} c_{t+k} \mid s_t, a_t\right], \tag{2}$$

where $\gamma_{\text{risk}} \in [0, 1)$ is the risk discount factor. This formulation assigns high risk not only to actions that are immediately unsafe, but also to actions that may lead to future constraint violations. The corresponding Bellman backup is

$$\begin{aligned} Q^{\pi}_{\text{risk}}(s_t, a_t) = c_t + (1 - c_t)\gamma_{\text{risk}} \\ \times \mathbb{E}_{a_{t+1}\sim\pi(\cdot|s_{t+1})}\left[Q^{\pi}_{\text{risk}}(s_{t+1}, a_{t+1})\right]. \end{aligned} \tag{3}$$

The factor $(1 - c_t)$ treats violations as absorbing and propagates failure information backward along trajectories. In principle, safe policy learning can be viewed as maximizing task return while keeping predicted risk below a tolerance,

$$Q^{\pi}_{\text{risk}}(s_t, a_t) \le \epsilon_{\text{risk}}. \tag{4}$$

Although this risk constraint provides a natural formulation for safe RL, we do not optimize it through direct constrained policy optimization. Instead, we formulate safety as an execution-time filtering problem, where each action proposed by π_{task} is either executed or replaced by recovery behavior. The filtering decision depends on both the predicted future risk and the reliability of that prediction under limited data and distribution shift.

IV. Method

We introduce Safe Evidential Real-World Reinforcement Learning (SEVI), a modular framework for uncertainty-aware safety intervention during online real-world RL. As shown in Fig. 2, SEVI combines three components: an Evidential Safety Critic for risk and uncertainty estimation, an offline-trained recovery policy for corrective intervention, and a task policy trained with sample-efficient off-policy RL using prior data. At runtime, each action proposed by the task policy is evaluated before execution. If the predicted risk or epistemic uncertainty exceeds a threshold, the proposed action is replaced by a recovery action. This design separates task learning from safety intervention while modifying only the executed action.

A. Evidential Safety Critic (ESC)

The ESC provides a reliability-aware safety estimate for execution-time intervention. It predicts a posterior predictive distribution over the temporal-difference risk target, from which we obtain both the expected future constraint-violation risk and epistemic uncertainty in a single forward pass. This enables the safety layer to reject actions that are either high-risk or poorly supported by evidence.

For a transition (s_t, a_t, s_{t+1}, c_t), we define the risk Bellman target as

$$y_t = c_t + (1 - c_t)\gamma_{\text{risk}}\mathbb{E}_{a_{t+1}\sim\pi(\cdot|s_{t+1})}\left[\hat{Q}^{-}_{\text{risk}}(s_{t+1}, a_{t+1})\right], \tag{5}$$

where $c_t \in \{0, 1\}$ is the safety-violation label and $\gamma_{\text{risk}} \in [0, 1)$ is the risk discount factor. If $c_t = 1$, the factor $(1 - c_t)$ treats safety violations as absorbing, preventing the target from bootstrapping beyond a violation. The target prediction $\hat{Q}^{-}_{\text{risk}}$ is computed using a slowly updated target critic with parameters $\bar{\phi}$, following standard temporal-difference learning. Since $y_t \in [0, 1]$ is a bounded violation-probability target, we train the evidential critic in logit space. We first clip the Bellman target for numerical stability and define

$$\bar{y}_t = \text{clip}(y_t, \epsilon, 1 - \epsilon), \qquad z_t = \text{logit}(\bar{y}_t). \tag{6}$$

The transformed target z_t is modeled as a noisy scalar observation generated from a Gaussian likelihood with unknown mean and variance,

$$z_t \sim \mathcal{N}(\mu_t, \sigma_t^2). \tag{7}$$

Following deep evidential regression [9], we place a Normal–Inverse–Gamma (NIG) prior over the unknown likelihood parameters (μ_t, σ_t^2):

$$\mu_t \sim \mathcal{N}(m_t, \sigma_t^2 \nu_t^{-1}), \qquad \sigma_t^2 \sim \Gamma^{-1}(\alpha_t, \beta_t). \tag{8}$$

For each state–action pair, the critic network outputs the corresponding evidential parameters,

$$(m_t, \nu_t, \alpha_t, \beta_t) = f_\phi(s_t, a_t), \tag{9}$$

where m_t is the predictive mean in logit space. The remaining three parameters define the evidential distribution over the unknown likelihood parameters. Specifically, ν_t controls the strength of evidence supporting the mean estimate m_t, α_t is the shape parameter of the inverse-gamma distribution over the variance, and β_t is its scale parameter. The predicted future violation risk is obtained by mapping the expected latent mean back to probability space:

$$\hat{Q}_{\text{risk}}(s_t, a_t) = \sigma(m_t), \tag{10}$$

where $\sigma(\cdot)$ denotes the sigmoid function. The target critic uses the same mapping,

$$\hat{Q}^{-}_{\text{risk}}(s_t, a_t) = \sigma(m_{\bar{\phi}}(s_t, a_t)), \tag{11}$$

where $m_{\bar{\phi}}$ is the predictive mean produced by the target network. Marginalizing over the unknown likelihood parameters (μ_t, σ_t^2) gives the Student-t posterior predictive distribution,

$$p(z_t \mid m_t, \nu_t, \alpha_t, \beta_t) = \text{St}\left(z_t; m_t, \frac{\beta_t(1+\nu_t)}{\nu_t \alpha_t}, 2\alpha_t\right). \tag{12}$$

Here, m_t is the location parameter, $\frac{\beta_t(1+\nu_t)}{\nu_t\alpha_t}$ is the Student-t scale parameter, and $2\alpha_t$ is the degrees of freedom. The negative log-likelihood of this predictive distribution defines the evidential regression loss,

$$L_{\text{NIG}} = -\log p(z_t \mid m_t, \nu_t, \alpha_t, \beta_t). \tag{13}$$

To discourage overconfident incorrect predictions, we use the evidence regularizer

$$L_{\text{reg}} = |z_t - m_t|\,(2\nu_t + \alpha_t). \tag{14}$$

This term penalizes high evidence when the transformed target z_t is far from the predictive mean m_t, discouraging overconfident risk estimates in poorly modeled state–action regions. The final evidential TD loss is

$$L_{\text{ESC}}(s_t, a_t, s_{t+1}, c_t; \phi) = L_{\text{NIG}} + \lambda_{\text{evi}} L_{\text{reg}}, \tag{15}$$

where λ_{evi} controls the strength of the evidence regularizer. The critic parameters are optimized over replay data by minimizing

$$J^{\text{ESC}}_{\text{risk}}(\phi) = \mathbb{E}_{(s_t, a_t, s_{t+1}, c_t)\sim\mathcal{D}}\left[L_{\text{ESC}}(s_t, a_t, s_{t+1}, c_t; \phi)\right]. \tag{16}$$

The NIG distribution provides a closed-form estimate of epistemic uncertainty. Since epistemic uncertainty corresponds to uncertainty over the latent predictive mean μ_t, we compute

$$\sigma^2_{\text{epi}}(s_t, a_t) = \text{Var}[\mu_t] = \frac{\beta_t}{\nu_t(\alpha_t - 1)}. \tag{17}$$

A large value of $\sigma^2_{\text{epi}}(s_t, a_t)$ indicates that the critic has limited evidence for the current state–action pair, which can occur in underexplored or out-of-distribution regions. Thus, $\hat{Q}_{\text{risk}}(s_t, a_t)$ estimates the likelihood of future constraint violation, while $\sigma^2_{\text{epi}}(s_t, a_t)$ measures the reliability of that estimate. Both quantities are passed to the recovery gate for uncertainty-aware action filtering.

B. Recovery Policy

The recovery policy π_{rec} provides a fallback behavior when the task policy proposes an action that is either high-risk or insufficiently supported by the safety critic. At each timestep, the task policy first proposes an action $a_t^{\text{task}} \sim \pi_{\text{task}}(\cdot \mid s_t)$. The ESC then evaluates the pair (s_t, a_t^{task}) using the predicted risk $\hat{Q}_{\text{risk}}(s_t, a_t^{\text{task}})$ and epistemic uncertainty $\sigma^2_{\text{epi}}(s_t, a_t^{\text{task}})$. We define the recovery set as

$$\mathcal{T}_{\text{rec}} = \left\{(s,a) \,\middle|\, \hat{Q}_{\text{risk}}(s,a) \geq \tau_{\text{risk}} \;\text{ or }\; \sigma^2_{\text{epi}}(s,a) \geq \tau_{\text{epi}}\right\}, \tag{18}$$

where τ_{risk} and τ_{epi} are safety and uncertainty thresholds calibrated on held-out offline data. If the proposed task action lies outside this set, it is executed. Otherwise, the action is replaced by the recovery policy:

$$a_t = \begin{cases} a_t^{\text{task}}, & (s_t, a_t^{\text{task}}) \notin \mathcal{T}_{\text{rec}}, \\ a_t^{\text{rec}}, & (s_t, a_t^{\text{task}}) \in \mathcal{T}_{\text{rec}}, \end{cases} \tag{19}$$

where $a_t^{\text{rec}} \sim \pi_{\text{rec}}(\cdot \mid s_t)$. This rule leaves the task-policy objective unchanged and applies safety intervention only through execution-time action replacement. An ESC-only filter would require resampling or selecting another admissible task action, whereas π_{rec} provides an explicit corrective action that moves execution away from failure-prone regions before returning control to π_{task}. Learning recovery behavior online is undesirable on physical robots because recovery actions are needed near failure boundaries, where exploratory behavior can cause damage or require human intervention. We therefore train π_{rec} offline using a dataset of human teleoperated recovery demonstrations $\mathcal{D}_{\text{rec}}$, collected from near-failure states back toward low-risk operating regions. To align recovery with the learned safety model, we relabel recovery data using an ESC-shaped reward based on predicted risk:

$$r_t^{\text{rec}} = -\hat{Q}_{\text{risk}}(s_t, a_t). \tag{20}$$

The recovery reward in Eq. (20) depends only on the predicted mean risk and does not incorporate epistemic uncertainty. Epistemic uncertainty is used exclusively by the recovery gate to determine when recovery is activated, thereby separating the recovery objective from the uncertainty-aware intervention criterion. We train π_{rec} with TD3+BC [14] on $\mathcal{D}_{\text{rec}}$. The value-based objective encourages low-risk recovery behavior, while behavior cloning regularization keeps the learned policy close

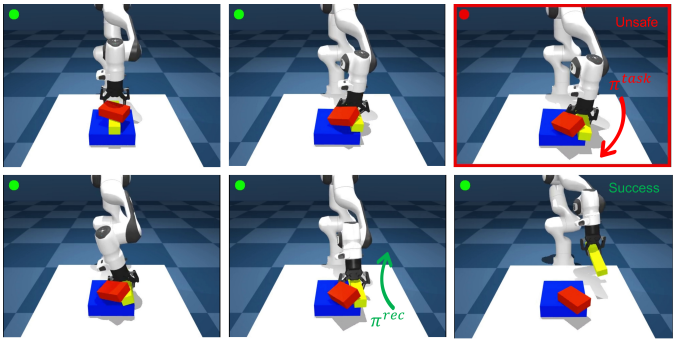


Fig. 3: Block Plucking example of SEVI task–recovery switching. An unsafe π^{task} action that could destabilize the stack (red) triggers π^{rec}, which returns the robot to a safer configuration before task execution resumes successfully (green).

to demonstrated corrective actions and reduces extrapolation outside the recovery dataset. During online real-world RL, the recovery policy acts as a stabilizing fallback. It is invoked only when the proposed task action enters the recovery set, and control returns to the task policy once the proposed task actions satisfy the risk and uncertainty thresholds.

C. Integration with Online Real-World RL

In our implementation, the task policy π_{task} is trained with RLPD within the SERL framework [2]. The task learner uses prior demonstrations, off-policy actor-critic updates, and online replay in the standard SERL pipeline. SEVI leaves the task reward, actor objective, critic update, and replay sampling procedure unchanged. At each timestep, the task policy proposes an action $a_t^{\text{task}} \sim \pi_{\text{task}}(\cdot \mid s_t)$. The executed transition is stored in the online replay buffer and reused for task-policy updates. Thus, the task policy continues to optimize the original task objective, while the collected data are biased toward lower-risk interactions. SEVI is intended as a practical execution-time safety layer, similar in spirit to Recovery RL and runtime shielding. Since both the safety critic and recovery policy are learned from finite data, SEVI does not provide formal guarantees of zero violations. Instead, its goal is to reduce unsafe interactions during online real-world RL while preserving the sample efficiency and task-learning behavior of the underlying RL algorithm.

V. Results

Our experiments are designed to assess three critical aspects of SEVI: **(Q1)** Can ESC provide reliable safety prediction under distribution shift with real-time inference? **(Q2)** Does epistemic uncertainty improve recovery gating beyond predicted risk alone? **(Q3)** Can SEVI reduce unsafe interactions during online real-world RL while preserving task performance?

We first evaluate safety prediction under in-distribution (ID) and out-of-distribution (OOD) conditions in both simulation and hardware to address **(Q1)** in Sec. V-B. We then examine **(Q2)** in Sec. V-C by analyzing both the quality of epistemic uncertainty and its effect on risk-based recovery gating. Finally, we address **(Q3)** in Sec. V-D through closed-loop online RL experiments in simulation and on a real robot, evaluating whether SEVI improves safety without compromising task success.

A. Evaluation Setup

Baselines. We evaluate SEVI with two groups of baselines. The first group measures closed-loop safe task learning and includes SERL [2], SAC [15], SAC-C [10], SQRL-style recovery [4], [16], SEVI without uncertainty, and full SEVI. SERL and SAC evaluate task learning without execution-time recovery filtering, while SAC-C evaluates constraint-penalized policy learning. The SQRL-style recovery baseline uses a point-estimate safety critic trained with the SQRL future-violation risk objective, but uses the same offline-trained recovery policy as SEVI. This keeps the recovery behavior fixed and isolates the effect of the safety critic used for intervention. SEVI without uncertainty uses the evidential critic but triggers recovery only from predicted mean risk, while full SEVI triggers recovery using both predicted risk and epistemic uncertainty. The second group evaluates safety prediction and uncertainty estimation independently of closed-loop task performance. We compare the ESC against a point-estimate SQRL-style critic [16], MC Dropout [8], and Deep Ensembles [7]. These baselines use comparable observation-action inputs and safety-label supervision, but differ in how prediction reliability is represented. This allows us to separately evaluate the closed-loop benefit of uncertainty-aware recovery and the standalone quality of the proposed safety critic.

Platforms and Environments. We evaluate SEVI on two visual manipulation tasks in MuJoCo and one hardware task using a Franka Research 3 arm. In the **Block Plucking** task, the robot must remove the middle block from a three-block stack while keeping the top block supported by the bottom block as shown in Fig. IV-C. The policy observes two 128×128 RGB views from front and wrist cameras together with a 16D state consisting of the 3D TCP position, 3D TCP linear velocity, gripper open/close state, and the 3D Cartesian positions of the three cubes; each episode runs at 50 Hz for a 300-step horizon. In **Transport Nut**, the robot must transport a square nut and insert it onto a target peg. Safety violations are defined as intervention-requiring events, including dropping the nut, moving it outside the reachable workspace, or causing insertion jamming due to large pose misalignment. For hardware evaluation, the **USB Insertion** task requires the robot to pick up a USB connector and insert it into a target slot while avoiding keep-out regions on the motherboard and excessive contact forces. The hardware policy observes dual 128×128 RGB views with a 19D state consisting of the 6D TCP pose (position and orientation), 6D TCP velocity (linear and angular), gripper open/close state, 3D TCP force, and 3D TCP torque, and executes at 10 Hz for a 120-step horizon. Distribution shift is evaluated through visual and task-configuration perturbations in simulation, and through distractor objects and cable-layout variations in the USB Task. All simulation training and real-robot experiments are run on a workstation with an Intel Core i9-14900KF CPU, 64 GB RAM, and an NVIDIA RTX 4090 GPU with 24 GB memory, running Ubuntu 22.04.5 LTS.

Evaluation Metrics and Protocol. In simulation, each method is trained with 5 independent random seeds, and the

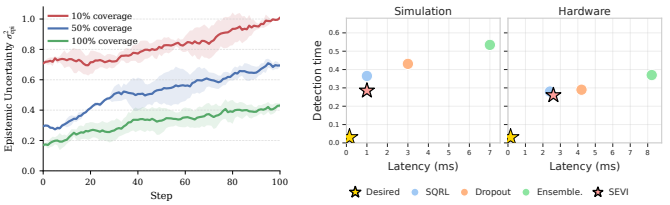
final policy from each seed is evaluated over 20 rollouts, yielding 100 evaluation rollouts per method for each task. For hardware evaluation, each method is trained in three independent 3-hour real-world online RL sessions on the USB Insertion task, corresponding to 9 hours of hardware training per method. For each run, the final policy is evaluated over 50 hardware rollouts, while unsafe events are counted over the entire online training session. Following real-world RL evaluation practice, we report the success rate of the final policy over 50 evaluation rollouts, while unsafe events and recovery interventions are counted over the entire online training session. This separates final task performance from the safety cost incurred during real-world policy improvement. We report three groups of metrics. Following prior runtime failure-prediction protocols [13], [17], we report trajectory-level TPR, TNR, balanced accuracy (BA), normalized detection time (DT), and inference latency. BA is computed for each seed as $\mathrm{BA} = \frac{1}{2}(\mathrm{TPR} + \mathrm{TNR})$, and then reported as the mean and standard deviation across seeds. For uncertainty quality, we report confidence AUROC/APR, OOD AUROC/APR, expected calibration error (ECE), and Brier score [18]. For closed-loop execution, we report task success rate, total unsafe events during training, recovery intervention rate, and cycle time. Closed-loop simulation and hardware results are reported as mean and standard deviation across five and three independent runs, respectively.

Implementation Details. The implementation details of the proposed modules are as follows:

Evidential Safety Critic (ESC): The ESC uses a twin-critic architecture to reduce overconfident risk estimates. Dual-camera RGB observations from wrist and front views are processed by a frozen ResNet-v1-10 encoder with shared weights. The visual embeddings are concatenated with a linear projection of the proprioceptive state and action. The fused representation is passed through a two-layer MLP with 256 hidden units, Swish activations, and LayerNorm, followed by an evidential output head that predicts the NIG parameters (m, ν, α, β). For conservative target computation and action filtering, we take the maximum predicted risk and maximum epistemic uncertainty across the two critics. The thresholds τ_{risk} and τ_{epi} are calibrated once before online RL using 50 successful and 50 failure validation rollouts and kept fixed thereafter. The calibrated pairs are Block Plucking $(0.44, 0.56)$, Transport Nut $(0.61, 0.64)$, and USB Insertion $(0.65, 0.45)$. Perturbing both thresholds by $\pm 20\%$ changes the success rate by less than 8%, 9%, and 5%, respectively. Safety labels c_t are obtained from privileged simulation states or hardware failure annotations.

Recovery Policy (π_{rec}): The recovery policy is trained offline with TD3+BC on teleoperated recovery trajectories collected near task-specific failure boundaries. We use 150 recovery trajectories for each simulation task and 200 trajectories for the hardware task. The actor and critic use two-layer MLPs with 256 hidden units, learning rate 1×10^{-4}, discount factor $\gamma = 0.95$, and behavior cloning weight $\lambda_{\mathrm{BC}} = 2.5$. The same recovery policy is used across recovery-based variants to isolate the effect of the safety critic and gating rule.

Task Policy (π_{task}): The task policy is trained with the SERL/RLPD off-policy RL pipeline using 50 successful demonstrations as prior data. The actor and critic use two-layer MLPs with 256 hidden units, learning rate 3×10^{-4}, a 200k-transition replay buffer, update-to-data ratio of 4, and automatic entropy tuning with target entropy $-\dim(\mathcal{A})/2$.



(a) (b)

Fig. 4: Characterization of the Evidential Safety Critic. (a) Epistemic uncertainty under increasing training coverage in a fixed USB failure region; shading denotes ± 1 standard deviation across three ESC trainings. (b) Detection efficiency of safety critics in simulation and hardware.

TABLE I: Trajectory-level safety prediction under ID and OOD conditions. SEVI achieves the highest balanced accuracy across tasks with consistent TPR–TNR balance.

Task	Metric	SQRL	Dropout	Ensemble	SEVI(w/o Unc.)	**SEVI**
Block ID	BA↑	.85±.01	.82±.03	.90±.03	.83±.04	**.93±.03**
	TPR↑	.77±.03	.74±.04	.86±.04	.80±.05	**.92±.03**
	TNR↑	.93±.03	.90±.03	**.95±.02**	.86±.04	.94±.04
Block OOD	BA↑	.66±.03	.69±.01	.74±.05	.66±.03	**.80±.02**
	TPR↑	.61±.03	.54±.03	.63±.07	.61±.04	**.73±.03**
	TNR↑	.71±.03	.84±.03	.85±.04	.72±.03	**.88±.03**
Nut ID	BA↑	.52±.02	.59±.04	.64±.04	.52±.03	**.75±.03**
	TPR↑	.23±.02	.31±.06	.39±.05	.27±.03	**.66±.04**
	TNR↑	.81±.03	.86±.04	**.89±.05**	.77±.05	.83±.04
Nut OOD	BA↑	.49±.03	.54±.03	.59±.03	.46±.02	**.65±.01**
	TPR↑	.38±.02	**.55±.04**	.34±.02	.38±.03	.48±.03
	TNR↑	.59±.05	.54±.04	**.85±.05**	.54±.04	.83±.03
USB ID	BA↑	.70±.02	.73±.01	.79±.05	.66±.04	**.84±.03**
	TPR↑	.58±.04	.67±.02	.82±.07	.52±.05	**.82±.04**
	TNR↑	.81±.07	.79±.02	.75±.04	.80±.05	**.86±.04**
USB OOD	BA↑	.58±.03	.58±.05	.64±.03	.59±.03	**.74±.02**
	TPR↑	.51±.06	.40±.05	.43±.05	.54±.05	**.57±.02**
	TNR↑	.65±.03	.77±.05	.85±.02	.64±.04	**.91±.03**

B. Safety Prediction under Distribution Shift

We first evaluate the safety critic independently of closed-loop task learning. Table I reports trajectory-level safety prediction results across Block Plucking, Transport Nut, and USB Insertion under both in-distribution (ID) and out-of-distribution (OOD) conditions. SEVI achieves the highest balanced accuracy in all evaluated settings, improving over the point-estimate SQRL and MC Dropout, while remaining comparable to or better than Deep Ensembles. This is important because ensembles are a strong uncertainty baseline, but require multiple independently trained critics, whereas SEVI obtains both risk and epistemic uncertainty in a single forward pass. The gains are most pronounced under distribution shift. On Block Plucking OOD, SEVI improves balanced accuracy from 0.66 for the SQRL-style critic to 0.80, and on USB Insertion OOD from 0.58 to 0.74. Transport Nut remains the most difficult setting because failures can occur across grasping, transport, and insertion phases. Nevertheless, SEVI

TABLE II: Detection time (DT) and inference latency of the evaluated safety critics corresponding to Fig. 4(b).

Method	Simulation (Block Plucking)		Hardware (USB Insert)	
	DT↓	Lat. (ms)↓	DT↓	Lat. (ms)↓
SQRL	0.37±0.03	1.00±0.08	0.28±0.02	2.40±0.15
Dropout	0.43±0.04	3.00±0.20	0.29±0.03	4.21±0.25
Ensemble	0.53±0.05	7.00±0.40	0.37±0.04	8.23±0.50
SEVI	**0.29±0.02**	**1.00±0.05**	**0.26±0.02**	**2.60±0.10**

TABLE III: Uncertainty estimation quality on the hardware task under ID and OOD conditions. SEVI provides stronger discrimination and calibration than baselines.

Set.	Metric	Dropout	Ensemble	**SEVI**
ID	AUROC ↑	0.68	0.88	**0.92**
	APR ↑	0.74	0.87	**0.96**
	ECE ↓	0.29	0.38	**0.07**
	Brier ↓	0.29	0.28	**0.04**
OOD	AUROC ↑	0.62	0.89	**0.94**
	APR ↑	0.25	0.21	**0.39**
	ECE ↓	0.13	0.30	**0.09**
	Brier ↓	0.12	0.19	**0.07**

achieves the highest OOD balanced accuracy. The TPR and TNR values show that this improvement is not caused by uniformly conservative predictions. SEVI improves unsafe-trajectory detection while maintaining strong safe-trajectory rejection, which is necessary for online RL where excessive false positives can suppress useful exploration.

Fig. 4(b) and Table II show that SEVI achieves the lowest normalized detection time on both Block Plucking and USB Insertion while maintaining latency comparable to the point-estimate SQRL critic. Unlike SQRL, however, SEVI also provides epistemic uncertainty in the same forward pass; MC Dropout and Deep Ensembles obtain uncertainty estimates at substantially higher inference cost. Together with Table I, these results show that SEVI provides accurate safety prediction, early detection, and low-latency uncertainty estimation, addressing Q1.

C. Uncertainty-Aware Recovery Gating

To answer Q2, we evaluate whether epistemic uncertainty is well calibrated, sensitive to training-data coverage, and useful for downstream recovery decisions. Table III reports uncertainty estimation results on the USB insertion task. SEVI achieves the strongest performance under both ID and OOD settings, obtaining higher AUROC/APR and lower ECE/Brier score than dropout and ensemble baselines. These results indicate that the evidential critic produces uncertainty estimates that are both discriminative and well calibrated under distribution shift. To assess whether epistemic uncertainty reflects limited data support, we train the ESC using 10%, 50%, and 100% of 200 trajectories collected from a fixed USB failure region and evaluate all models on the same 50 held-out trajectories. As shown in Fig. 4(a), epistemic uncertainty decreases consistently as training coverage increases, providing empirical evidence that the estimate captures limited support in the training data. Fig. 5 evaluates whether this uncertainty improves downstream action filtering. We compare three SEVI gating variants: risk-only, uncertainty-only, and risk-plus-uncertainty. The combined gate achieves the best

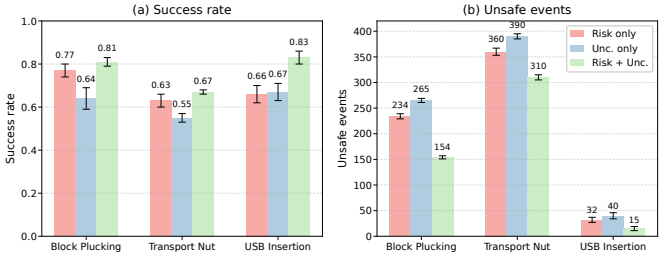


Fig. 5: Effect of recovery-gating criteria on closed-loop execution. Combining risk and uncertainty gives the best safety–performance trade-off across tasks.

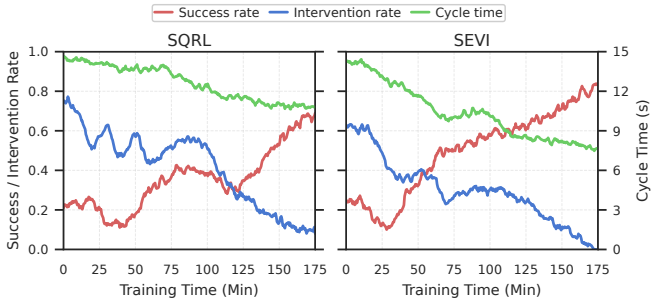


Fig. 6: Learning curves from a single representative hardware training run on the USB insertion task, showing success rate, intervention rate, and cycle time for SQRL and SEVI during online training. SEVI reaches higher success while progressively reducing intervention rate and cycle time.

safety–performance trade-off across all tasks. Relative to risk-only gating, it improves success by 4 percentage points on Block Plucking and 17 percentage points on USB Insertion, while reducing unsafe events by 34.2% and 53.1%, respectively. On Transport Nut, the combined gate reduces unsafe events by 13.9%. Aggregated across tasks, risk-plus-uncertainty gating reduces unsafe events by **23.5**% compared with risk-only gating and by **31.1**% compared with uncertainty-only gating. To assess whether this benefit depends on the recovery behavior, Table IV evaluates two qualitatively distinct USB recovery responses: Hit-and-Recover and Stuck-and-Recover. Risk-plus-uncertainty gating consistently improves task success and reduces unsafe events under both behaviors, indicating that the gating advantage persists across distinct recovery responses. These results demonstrate that predicted risk and epistemic uncertainty provide complementary intervention signals. Predicted risk captures the likelihood of future constraint violations, whereas epistemic uncertainty identifies state–action regions in which the safety estimate is poorly supported by data. Their combination therefore enables more selective recovery intervention while preserving task performance.

D. Safe Online Policy Learning

To address Q3, we evaluate whether SEVI reduces unsafe interactions during online RL without slowing policy learning or overly restricting execution. Fig. 6 shows that SEVI achieves this balance more effectively than SQRL under the fixed interaction budget on USB insertion task. After 175 minutes, SEVI reaches 0.84 success with a 7.6 s cycle time,

TABLE IV: Effect of recovery behavior on USB Insertion task.

Recovery Behavior	Gating	Success↑	Unsafe↓
Hit-and-Recover	Risk only	0.64 ± 0.03	32 ± 4
	Risk+Unc.	$\mathbf{0.82 \pm 0.03}$	$\mathbf{16 \pm 4}$
Stuck-and-Recover	Risk only	0.64 ± 0.01	33 ± 4
	Risk+Unc.	$\mathbf{0.83 \pm 0.03}$	$\mathbf{15 \pm 5}$

TABLE V: Closed-loop task execution performance. Success is evaluated using the final policy, while unsafe events are counted throughout training.

Method	Block Plucking		Transport Nut		USB Insertion	
	Succ. ↑	Unsafe ↓	Succ. ↑	Unsafe ↓	Succ. ↑	Unsafe ↓
SERL [2]	0.67±.02	270±7	0.52±.01	450±11	0.71±.03	56±7
SAC [15]	0.63±.01	298±5	0.49±.01	467±9	0.61±.03	49±4
SAC-C [10]	0.71±.03	281±3	0.55±.02	410±7	0.70±.04	37±6
SQRL [4]	0.75±.02	220±5	0.61±.01	358±5	0.68±.04	35±3
SEVI(w/o Unc.)	0.77±.03	234±5	0.63±.03	360±7	0.66±.04	32±5
SEVI	**0.81±.02**	**154±3**	**0.67±.01**	**310±5**	**0.83±.03**	**15±4**

compared with 0.69 success and 11.4 s for SQRL. Repeated task–recovery switching can occur during early training when the task policy repeatedly proposes inadmissible actions; however, the declining intervention rate in Fig. 6 indicates that such switching becomes less frequent as the policy improves. In contrast, SQRL exhibits larger intervention fluctuations and slower success improvement, suggesting that mean-risk filtering alone provides a weaker intervention signal.

Table V reports closed-loop task performance across simulation and hardware. For USB Insertion, results are reported as mean and standard deviation across three independent hardware training runs. SEVI provides the strongest safety–performance trade-off, achieving the highest success rate and the lowest unsafe-event count on all three tasks. Relative to SQRL, SEVI reduces unsafe events by 30.0% on Block Plucking, 13.4% on Transport Nut, and 57.1% on USB Insertion. SERL and SAC achieve reasonable task performance but accumulate substantially more unsafe events due to the absence of execution-time recovery filtering. SAC-C reduces violations through constraint penalties, but remains less effective than SEVI, indicating that fixed penalties are insufficient for policy-induced safety risks in complex manipulation tasks. Overall, these results support Q3: reliability-aware recovery reduces unsafe interactions during online policy learning while preserving task improvement and execution efficiency.

VI. Conclusion

We presented Safe Evidential Real-World Reinforcement Learning (SEVI), a modular framework that decouples task-policy learning from execution-time safety intervention. SEVI combines an Evidential Safety Critic, which estimates future constraint-violation risk and epistemic uncertainty in a single forward pass, with an offline-trained recovery policy that intervenes when proposed actions are predicted to be unsafe or insufficiently supported by evidence. Across two simulated manipulation tasks and a real-world USB insertion task, SEVI improves safety prediction under distribution shift and reduces unsafe interactions while maintaining task performance. The current framework has several practical limitations. Recovery behavior is not explicitly optimized for local or motion-efficient correction, and repeated task–recovery switching may occur near failure boundaries during early stages of policy learning. In addition, the learned safety components do not provide formal guarantees of zero violations. Future work will focus on more efficient corrective behaviors, improved switching mechanisms, and tighter integration with formally constrained safety methods.

References

[1] P. J. Ball, L. Smith, I. Kostrikov, and S. Levine, "Efficient online reinforcement learning with offline data," 2023. [Online]. Available: https://arxiv.org/abs/2302.02948

[2] J. Luo, Z. Hu, C. Xu, Y. L. Tan, J. Berg, A. Sharma, S. Schaal, C. Finn, A. Gupta, and S. Levine, "Serl: A software suite for sample-efficient robotic reinforcement learning," 2024.

[3] H. Zhu, J. Yu, A. Gupta, D. Shah, K. Hartikainen, A. Singh, V. Kumar, and S. Levine, "The ingredients of real-world robotic reinforcement learning," 2020. [Online]. Available: https://arxiv.org/abs/2004.12570

[4] B. Thananjeyan, A. Balakrishna, S. Nair, M. Luo, K. Srinivasan, M. Hwang, J. E. Gonzalez, J. Ibarz, C. Finn, and K. Goldberg, "Recovery rl: Safe reinforcement learning with learned recovery zones," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4915–4922, 2021.

[5] J. Seo, K. Nakamura, and A. Bajcsy, "Uncertainty-aware latent safety filters for avoiding out-of-distribution failures," *Conference on Robot Learning (CoRL)*, 2025.

[6] K. Nakamura, L. Peters, and A. Bajcsy, "Generalizing safety beyond collision-avoidance via latent-space reachability analysis," in *Robotics: Science and Systems XXI*, ser. RSS2025. Robotics: Science and Systems Foundation, 2025. [Online]. Available: http://dx.doi.org/10.15607/RSS.2025.XXI.113

[7] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," 2017. [Online]. Available: https://arxiv.org/abs/1612.01474

[8] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," 2016. [Online]. Available: https://arxiv.org/abs/1506.02142

[9] A. Amini, W. Schwarting, A. P. Soleimany, and D. Rus, "Deep evidential regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[10] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

[11] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Transactions on Automatic Control*, vol. 64, no. 8, p. 334–349, 2019.

[12] M. Chen, S. Herbert, and C. J. Tomlin, "Hamilton–jacobi reachability: A brief overview and recent advances," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 6, p. 101–128, 2023.

[13] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. Ambrus, H. Nishimura, and M. Itkina, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," *arXiv preprint arXiv:2503.08558*, 2025.

[14] S. Fujimoto and S. S. Gu, "A minimalist approach to offline reinforcement learning," 2021. [Online]. Available: https://arxiv.org/abs/2106.06860

[15] T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha, J. Tan, V. Kumar, H. Zhu, A. Gupta, P. Abbeel, and S. Levine, "Soft actor-critic algorithms and applications," 2019. [Online]. Available: https://arxiv.org/abs/1812.05905

[16] K. Srinivasan, B. Eysenbach, S. Ha, J. Tan, and C. Finn, "Learning to be safe: Deep rl with a safety critic," 2020. [Online]. Available: https://arxiv.org/abs/2010.14603

[17] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, "Failure prediction at runtime for generative robot policies," 2025.

[18] M. Itkina and M. J. Kochenderfer, "Interpretable self-aware neural networks for robust trajectory prediction," 2022. [Online]. Available: https://arxiv.org/abs/2211.08701